Binary Search and Sorting

Topic 6 of 10

4 questions | skeleton and writing space only

Each question: the prompt, the function or class signature, and ruled lines to write your own solution.

1. Cutting Ribbons (LC 1891)

MEDIUM | asked 2026-08-16 | array, binary-search, complexity-analysis

LC 1891 Cutting Ribbons: given wood lengths and K, maximize the equal piece length that yields at least K pieces.

```
def max_length(ribbons, k):
```

2. Progressive Array Partitioning

MEDIUM | asked 2026-08-11 | array, hashmap, sorting

Examples: Stage 1 [1, 2, 3, 4, 2, 3, 3, 4, 4, 4] -> true because the four value-groups have different sizes. [1, 2, 3] -> false because every value-group has size one. Stage 2 [1, 2, 2, 3, 3, 4, 4, 5, 5, 6] -> true. [4, 5, 6, 7, 7, 8, 9] -> false.

```
def stage1(counts):
```

3. Streaming Points: Emit K Within Distance

MEDIUM | asked 2026-06-23 | data-structure, sorting, sliding-window, ambiguous-spec

Process a stream of float positions. The important ambiguity is whether the condition means a span of length d or pairwise distance within d.

```
class PointStream:

    def __init__(self, k, d):

    def window_start(self, pos):

    def add(self, x):
```

4. First Bad Version with Parallel Search Follow-up

MEDIUM | asked 2026-06-15 | binary-search, parallelism, concurrency

Phone-screen R1 coding round. The base problem is the standard first-bad-version monotonic search -- given an isBadVersion(v) oracle over versions 1..n, find the first bad version with a single binary search.

```
def first_bad_version(n, is_bad):
```